# Helix Forge Research Failure Analysis Research Brief

HELIX FORGE LAB / p04-full-005338 / RUN R184 / 2026-07-13

| ALPHA F1 | PRECISION | RECALL | BETA F1 |
|---|---|---|---|
| 0.873 | 0.881 | 0.865 | 0.846 |

## Decision context

This report records a bounded failure analysis evaluation on HF-Eval-S1/S2 v3.2. The 2,400-record population is synthetic. Alpha M-14 is retained as a research checkpoint, not as a deployment authorization.

## Engineering interpretation

Aggregate improvement remains meaningful only with the frozen scoring policy. The review therefore keeps slice-level calibration, dataset lineage, and the comparison with Beta B-07 attached to run R184.

## Lineage

Upstream evidence: p04-full-000055 (research/papers/2026-07-13/helix-forge-research/failure-analysis/tables/helix-forge-research-failure-analysis-thresholds-v01.csv).

| Slice | Records | Precision | Recall | F1 |
|---|---|---|---|---|
| slice-01 | 234 | 0.881 | 0.874 | 0.877 |
| slice-02 | 234 | 0.863 | 0.850 | 0.856 |
| slice-03 | 205 | 0.880 | 0.863 | 0.871 |
| slice-04 | 209 | 0.862 | 0.876 | 0.869 |
| slice-05 | 229 | 0.879 | 0.852 | 0.865 |